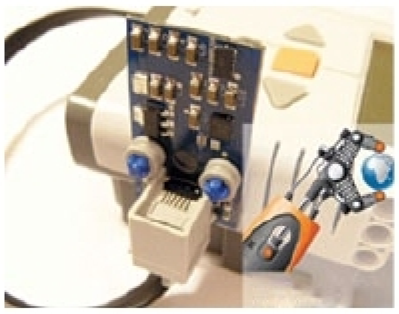

Fig. 8: Inertial sensor mounted on robot (Credit: www.dexterindustries.com)

Fig. 9: High-sensitivity sound sensor (Credit: http://fut-electronic.com)

Fig. 10: B5T HVC face-detection sensor module (Credit: www.mouser.com)

Fig. 11: RoboTouch Twendy-One gripper (Credit: https://pressureprofile.com)

tion and are wear resistant.

Inertial sensors. Motion sensing using MEMS inertial sensors is applied to a wide range of consumer products including computers, cell phones, digital cameras, gaming and robotics. There are various types of inertial sensors depending on applications. LORD MicroStrain manufactures industrial-grade inertial sensors that provide a wide range of tri-axial inertial measurements, and computed attitude and navigation solutions.

Sound sensors. Sound sensors are used in robots to receive voice commands. High-sensitivity microphones or sound sensors are essential in voice assistants like Google Assistant, Alexa and Siri.

Emotion sensors. Robots can react to human facial expressions using emotion sensors. B5T HVC face-detection sensor module from Omron Electronics is a fully-integrated human vision component (HVC) plug-in module that can identify faces with speed and accuracy. The module can evaluate the emotional mood based on one of the five programmed expressions.

Grip sensors. RoboTouch Twendy-One gripper features embedded digital output, which is ideal for OEM integration into a gripper. These sensors comprise multiple sensing pads, each having multiple sensing elements. Data from sensors is transferred to grippers via I2C or SPI digital interfaces, enabling easy integration of the sensors into grippers.

Some leading manufacturers of sensors for robotics

- Analog Devices
- Dexter Industries
- LORD MicroStrain
- Murata
- Omron
- Pressure Profile Systems Inc.
- Rockwell Automation
- Seeed Technology Co. Ltd
- Siemens
- Tamagawa Seiki

Artificial intelligence

Sensors used in AI robots are the same as, or are similar to, those used in other robots. Fully-functional human robots with AI algorithms require numerous sensors to simulate a variety of human and beyond-human capabilities. Sensors provide the ability to see, hear, touch and move like humans. These provide environmental feedback regarding surroundings and terrain.

Fig. 12: Atlas humanoid robot (Credit: https://en.wikipedia.org)

Distance, object detection, vision and proximity sensors are required for self-driving vehicles. These include camera, IR, sonar, ultrasound, radar and lidar.

A combination of various sensors allows an AI robot to determine size, identify an object and determine its distance.

Radio-frequency identification (RFID) are wireless sensor devices that provide identification codes and other information.

Force sensors provide the ability to pick up objects.

Torque sensors can measure and control rotational forces.

Temperature sensors are used to determine temperature and avoid potentially-harmful heat sources from the surroundings.

Microphones are acoustical sensors that help the robot receive voice commands and detect sound from the environment. Some AI algorithms can even allow the robot to interpret the emotions of the speaker.

Humanoid robots with AI algorithms can be useful for future distant space exploration missions. Atlas is a 183cm (6-feet) tall, bipedal humanoid robot, designed for a variety of search-and-rescue tasks for outdoor, rough terrains. It has an articulated sensor head that includes stereo cameras and a laser range finder.

To sum up

The right use of sensors can improve the performance of artificially-intelligent robots. Getting raw sensor data alone is not enough. AI tools can be useful when applied to sensor systems. Many new sensors are being developed, and the use of hybrid tools can further improve the strengths of AI robots.

New advances in machine intelligence are creating seamless interactions between people and digital sensor systems. AI has powerful tools for use along with sensor systems, for automatically solving problems that would normally require human intelligence. These are knowledge-based systems, fuzzy logic, learning, neural networks, reasoning and ambient intelligence. EFY